CAS Applied Data Science
Final Project

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BERN

Predicting Coffee Bean moisture content during roasting using Machine Learning

A critical examination of variable Informativeness

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Introduction

Coffee bean roasting is a critical process that significantly influences the quality, flavour, and physical properties of coffee beans. During roasting, moisture content changes dynamically as heat, environmental conditions, and processing parameters interact throughout the roasting cycle. Understanding and accurately predicting moisture behaviour is important for optimizing roasting consistency, improving product quality, and enhancing process control in coffee production. In this project, data collected from a coffee machine are environmental variables, processing conditions, and roasting parameters. These variables are used to analyse the factors affecting moisture content during roasting. Exploratory Data Analysis (EDA) and supervised machine learning techniques are applied to model moisture dynamics and evaluate predictive performance.

The main goals for the project are:

1. To evaluate and compare supervised machine learning models to identify the most robust approach for predicting coffee bean moisture content during roasting
2. To evaluate which factors, affect the moisture dynamics during roasting

Data

2.1 Data Provenance and Structure

The dataset used in this study was collected from a coffee roasting machine and consists of 9000 couples. The data captured various environmental conditions, processing parameters, and roasting characteristics throughout the coffee roasting process. A total of nine variables were recorded and used as predictor variables, while moisture content was considered the target variable for analysis. These variables were selected because they represent key operational factors that may influence moisture loss and moisture dynamics during roasting.

The purpose of collecting these variables is to investigate how different roasting conditions affect the movement and behaviour of moisture content within coffee beans. Factors such as water content, temperature, humidity, bean weight, group classification, and recipe settings may interact in complex ways during roasting. By analysing these variables, it is possible to identify patterns and relationships that contribute to changes in moisture content. Understanding these relationships is important for improving roasting consistency, optimizing process parameters, and enhancing overall coffee quality. The dataset therefore provides a valuable foundation for exploring the influence of multiple factors on moisture behaviour.

It is important to note that the dataset used in this project is synthetic and was generated for educational, experimental, and research purposes. The data do not represent actual commercial production records but were designed to simulate realistic roasting conditions and variable interactions. Despite being synthetic, the dataset enables the application of Exploratory Data Analysis (EDA) and machine learning techniques to investigate predictive modelling approaches and feature importance. This provides an opportunity to evaluate the effectiveness of different supervised learning models while developing a deeper understanding of the factors that influence moisture content during coffee bean roasting.

2.2 Data Flow

The data flow process began with downloading the dataset from Kaggle and importing it into a Visual Studio environment for further exploration and preprocessing. After loading the dataset, an initial inspection was conducted to understand the dataset structure, variable types, dimensions, and overall data quality. Descriptive statistics and visualizations were used to examine the distributions of variables and identify patterns from the results.

The dataset was prepared for analysis through preprocessing and cleaning procedures. Since the dataset was already cleaned and well-structured, no major corrections were required. The dataset did not contain any missing values, duplicated records, or inconsistent data entries, which simplified the preprocessing stage.

2.3 Data Quality

In this dataset, the overall data quality was considered normal. The dataset didn't have any missing values, which reduced the need for imputation methods or removal of incomplete records. Additionally, there were no duplicated entries detected, ensuring that each observation represented a unique couple.

The variables were also consistently formatted, allowing smooth integration into statistical analysis and machine learning workflows. Since the data is synthetic, there were fewer issues related to human error, inconsistent survey responses, or incomplete information that are commonly found in real-world datasets.

2.4 Feature Engineering and Preprocessing

Preprocessing was focused on preparing the variables and splitting them into Numerical and Categorical variables. Numerical variables were examined to ensure that they were within expected ranges and suitable for analysis. Categorical variables were examined to ensure that the categories were properly encoded and ready for Machine Learning.

Exploratory Data Analysis (EDA) techniques, such as histograms and distribution plots, were used to better understand the variables characteristics. These visualisations helped identify patterns, trends, and relationships within the data. The Moisture variable was analysed to evaluate the moisture behaviour across the different variables and to study the patterns and trends among them.

2.5 Data Cleaning

Minimal cleaning was required because the dataset was already pre-processed before analysis was being carried out by Nestle. There were no missing values or duplicate records in the dataset, which made the cleaning process easy. Nevertheless, the dataset was still reviewed carefully to verify consistency and ensure that all variables were correctly interpreted. Variable names, data types, and category labels were checked to confirm accuracy and compatibility with analysis tools.

Exploratory Data Analysis

Figure 1: Distribution of Moisture

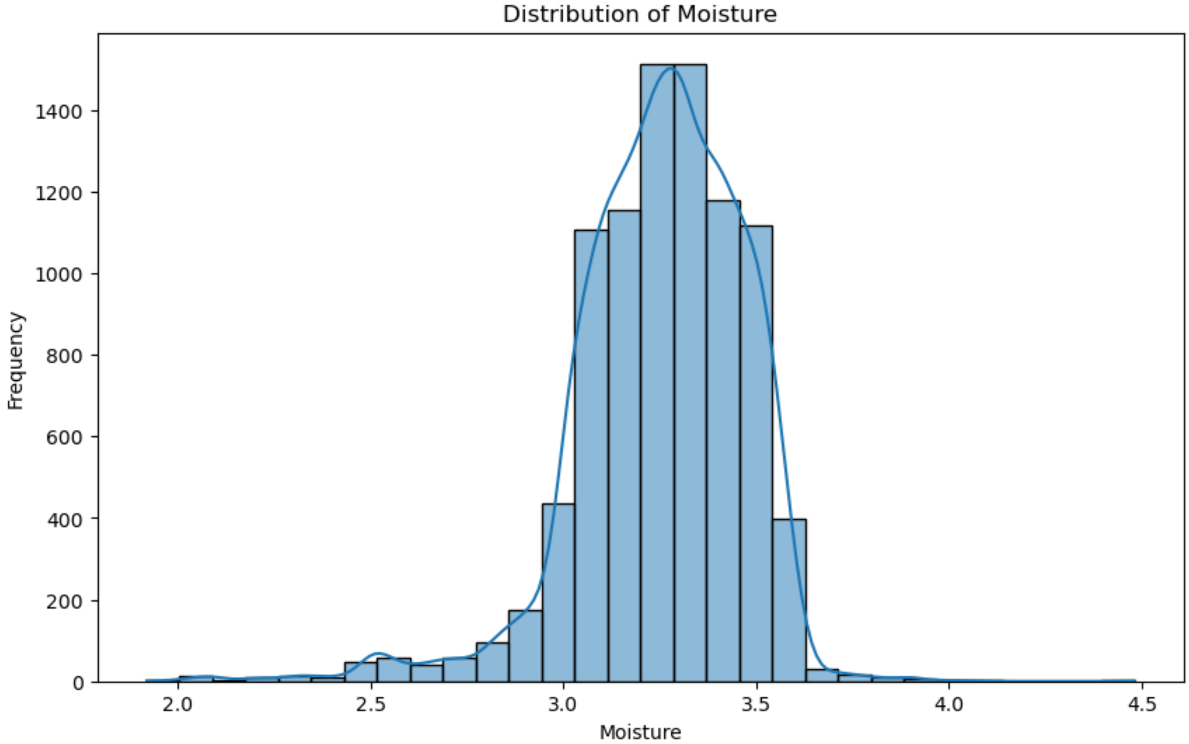


Figure 1 illustrates the distribution of moisture content across the dataset. The distribution appears relatively even, suggesting that the moisture values are spread consistently throughout the observations, which may indicate that the variables have a linear relationship between moisture content. The mean moisture content is 3.25, showing that, on average, the coffee beans retain a moisture level of 3.25 during the roasting process. In addition, the figure shows that the maximum observed moisture content is 3.25, while the minimum value is 1.92. This range highlights the variation in moisture levels throughout roasting and provides an initial indication of how moisture changes under different roasting conditions.

Figure 2: Numerical Variables

	Moisture1	Water	Humidity_environment	weight	Temperature_environment	Temperature1
count	9011.000000	9011.000000	9011.000000	9011.000000	9011.000000	9011.000000
mean	3.253889	45.174121	75.252928	288.287316	15.691703	206.132427
std	0.217269	2.058858	21.723469	6.321511	8.603261	68.084710
min	1.920000	30.000000	18.417000	54.000000	-5.420000	4.000000
25%	3.130000	44.100000	57.037500	289.000000	9.460000	224.900000
50%	3.270000	45.300000	79.875000	290.000000	14.770000	229.300000
75%	3.410000	46.500000	95.864000	291.000000	20.990000	231.100000
max	4.480000	58.000000	100.367000	458.000000	41.520000	237.800000

Figure 2 presents the distribution and summary of the numerical variables included in the dataset. From the figure, the numerical variables appear to be reasonably well spreaded across their respective ranges, which suggests sufficient variation for statistical analysis and machine learning modelling. A balanced spread is important because variables with little variation often contribute less predictive value, whereas variables with wider and more structured distributions may provide stronger signals for explaining changes in moisture content. The figure also helps identify whether the variables are symmetric, skewed, or concentrated around specific values, which is useful for understanding the underlying behaviour of the roasting process. Variables that appear evenly distributed may support more stable modelling, while variables with skewness or wider dispersion may indicate process fluctuations or stages in roasting where conditions change more rapidly. Overall, Figure 2 provides evidence that the numerical features contain meaningful variability and are suitable for further exploratory analysis, correlation assessment, and predictive modelling of coffee bean moisture content.

Figure 3: Boxplot of Moisture by Recipe

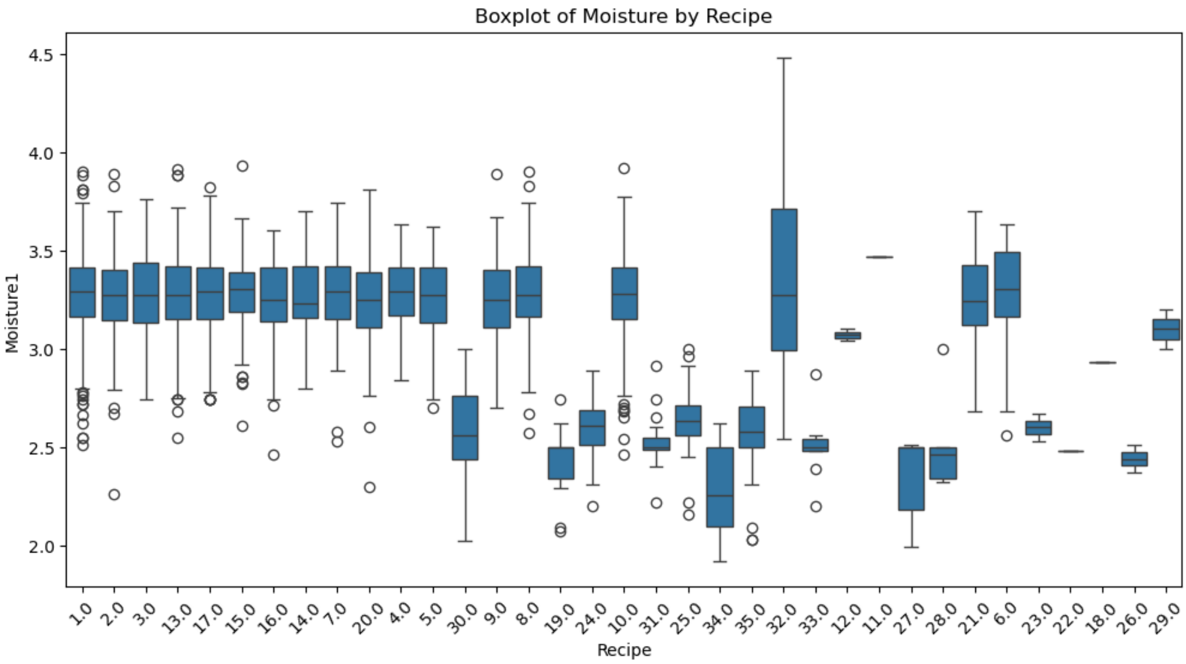


Figure 3 illustrates the distribution of moisture content across the different roasting recipes included in the dataset. The boxplots show variations in both the median moisture content and the spread of moisture values among recipes, indicating that recipe selection plays an important role in influencing moisture dynamics during the roasting process. Several recipes exhibit higher median moisture levels, while others are associated with lower moisture content, suggesting that differences in roasting parameters embedded within each recipe may affect the rate of moisture loss from the coffee beans.

In addition, the variability observed within certain recipes, as indicated by wider interquartile ranges and the presence of outliers, suggests that moisture content can fluctuate considerably under some roasting conditions. Recipes with narrower distributions demonstrate more consistent moisture levels and may provide greater process stability. These findings indicate that recipe configuration is a significant factor affecting moisture behaviour and support the inclusion of Recipe as an important predictor variable in the machine learning models developed for this study. The observed differences among recipes also suggest that roasting settings and operational conditions embedded within each recipe contribute to the overall moisture dynamics and should be considered when optimising roasting performance and moisture control.

Figure 4: Moisture by Batch Group

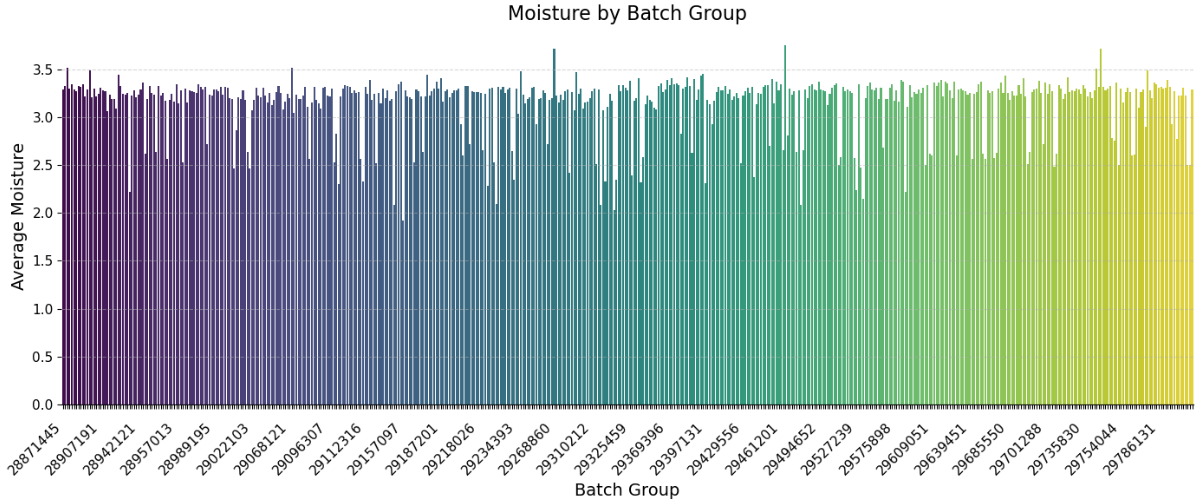


Figure 4 illustrates the batch groups and the average moisture content per batch groups. The moisture content varies based on the batch groups and this indicates that it's an important factor in predicting the moisture content during the roasting phase.

Although the differences in average moisture content across the batch groups are on a lower scale, a noticeable variation can still be observed. Most batch groups exhibit average moisture values ranging between approximately 2.0 and 3.5, with some batches showing slightly higher or lower moisture levels than others. This variation suggests that each batch may experience different roasting conditions, environmental influences, or processing characteristics that affect moisture retention and moisture loss during roasting. Factors such as water content, temperature conditions, humidity levels may contribute to these differences between batches.

The results also indicate that moisture content is not constant across all roasting batches, highlighting the dynamic nature of the roasting process. Since moisture content is a key indicator of roasting progression and bean quality, understanding batch-to-batch variability is important for achieving consistent roasting outcomes. The observed differences suggest that the batch group variable may capture underlying process characteristics that are not fully represented by the other measured variables in the dataset.

Figure 5: Time Dynamics of Moisture

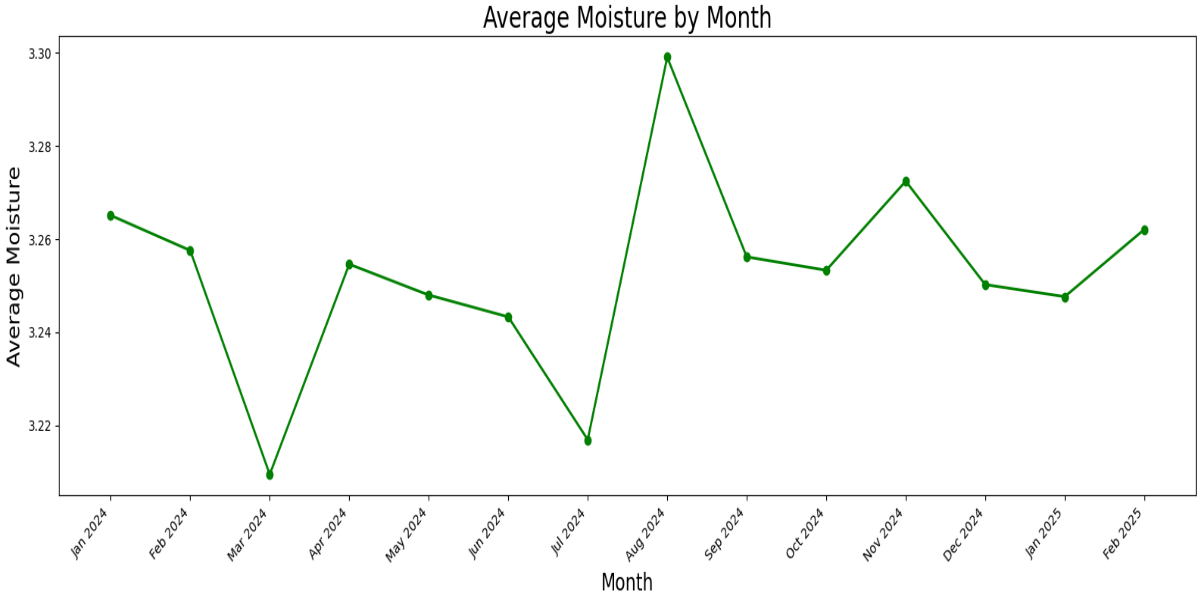


Figure 5 illustrates the moisture content across the 12 months. There is a trend that can be seen in this graph. The moisture content decreases in March, July, September and December indicating that the time could be the main driver that's causing the coffee bean to lose the moisture contents across the 12 months.

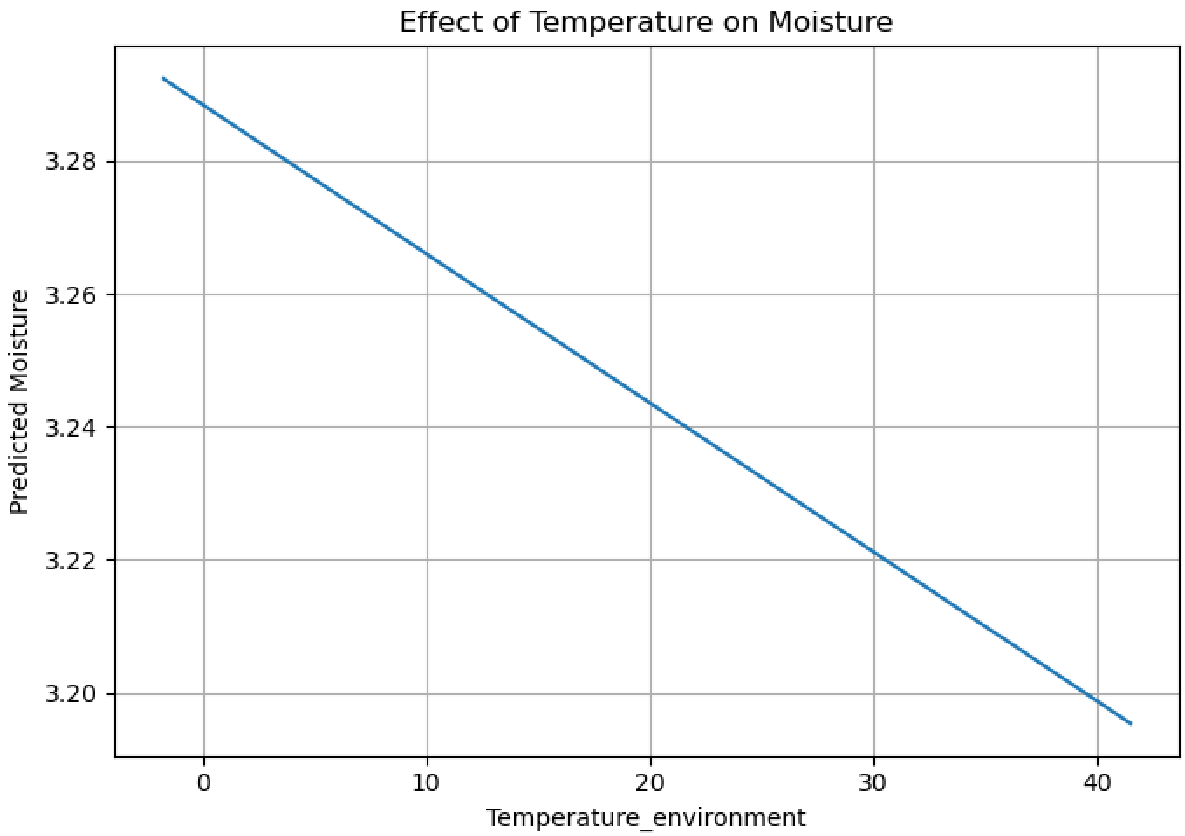
Although the fluctuations in average moisture content are small, the graph shows that moisture levels are not constant throughout the year. A decline can be observed in March and July, followed by an increase in August, which records the highest average moisture content during the study period. After August, moisture levels continue to fluctuate moderately before declining again towards the end of the year. These variations suggest that temporal factors may influence the moisture content of coffee beans during roasting.

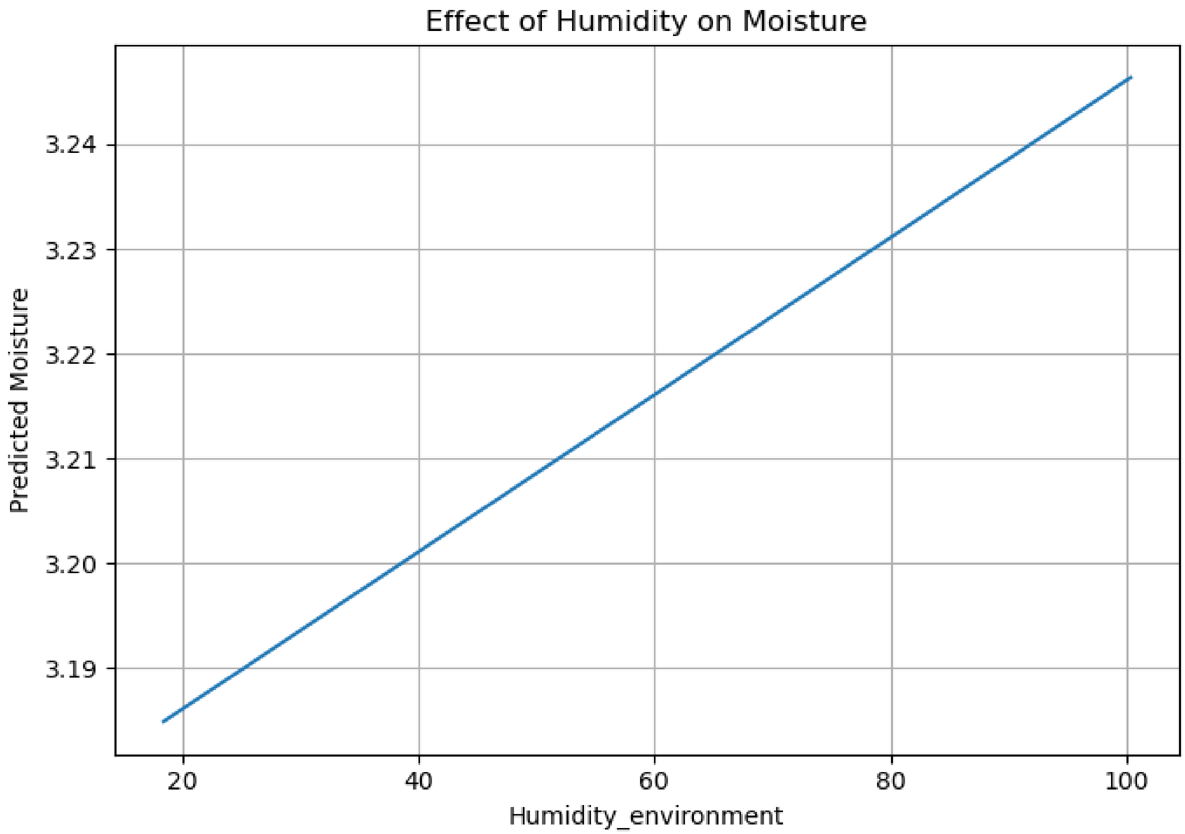
The observed pattern may be associated with seasonal changes in environmental conditions such as temperature and humidity. Variations in these environmental factors throughout the year can affect the initial moisture condition of the coffee beans as well as the rate at which moisture is lost during the roasting process. As a result, batches roasted in different months may exhibit different moisture characteristics even when similar roasting procedures are applied.

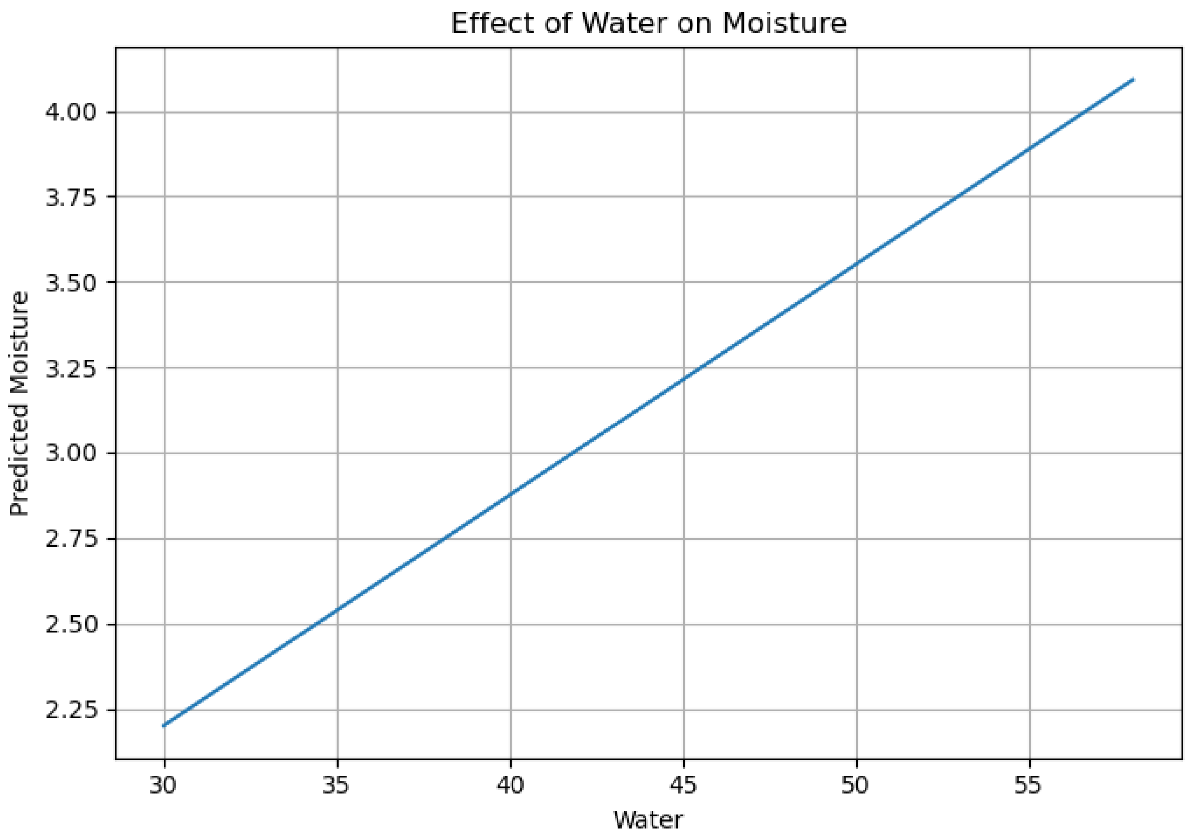
The graph also indicates that moisture content follows a dynamic pattern over time rather than remaining stable across all months. This suggests that time-related variables may contain useful information for predicting moisture content and should be considered in the machine learning models. While time alone may not directly cause moisture loss, it may serve as an indicator of underlying environmental and operational changes that occur throughout the year. Therefore, the monthly trend observed in Figure 5 highlights the importance of considering temporal effects when analysing moisture dynamics and developing predictive models for coffee bean roasting.

Machine Learning

Linear Regression




Effect of Humidity on Moisture
Predicted Moisture
3.24
3.23
3.22
3.21
3.20
3.19
20
40
60
80
100
Humidity_environment


Effect of Water on Moisture
Predicted Moisture
4.00
3.75
3.50
3.25
3.00
2.75
2.50
2.25
30
35
40
45
50
55
Water

Model Performance on Test Set:

```
Model Performance on Test Set:
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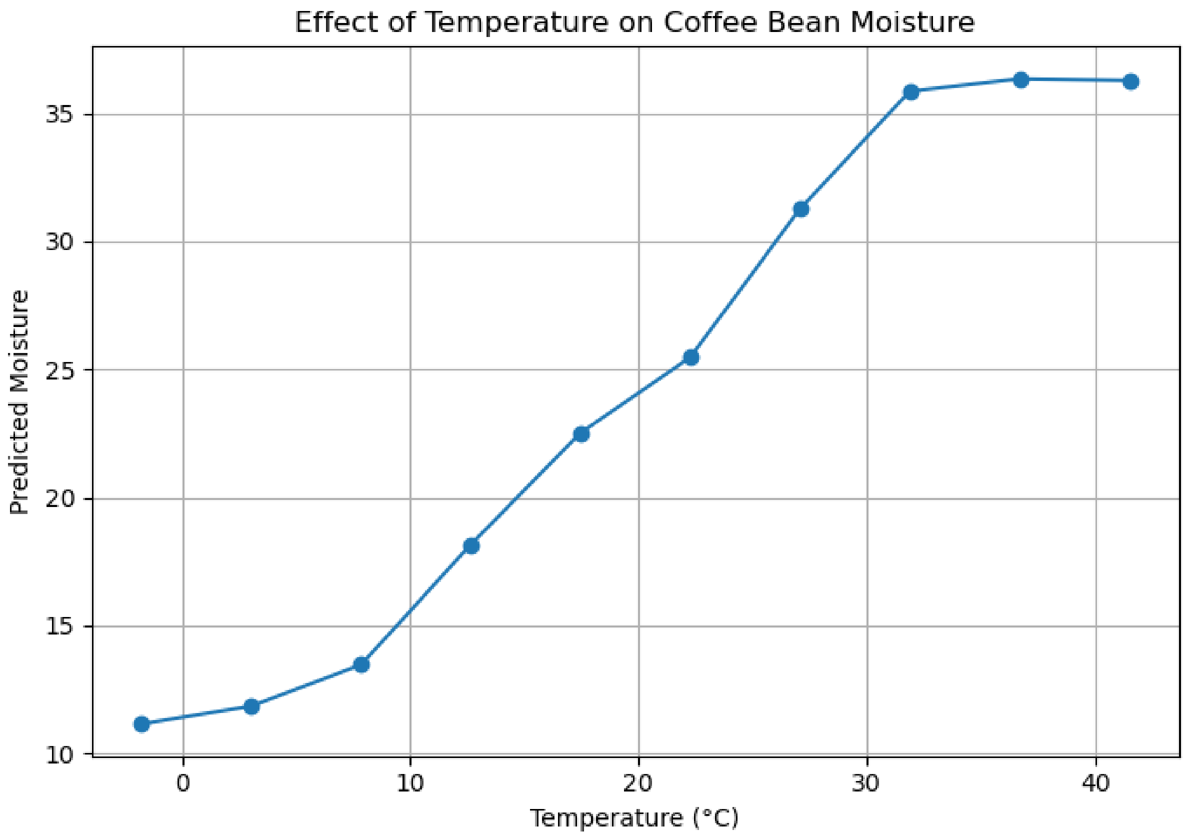
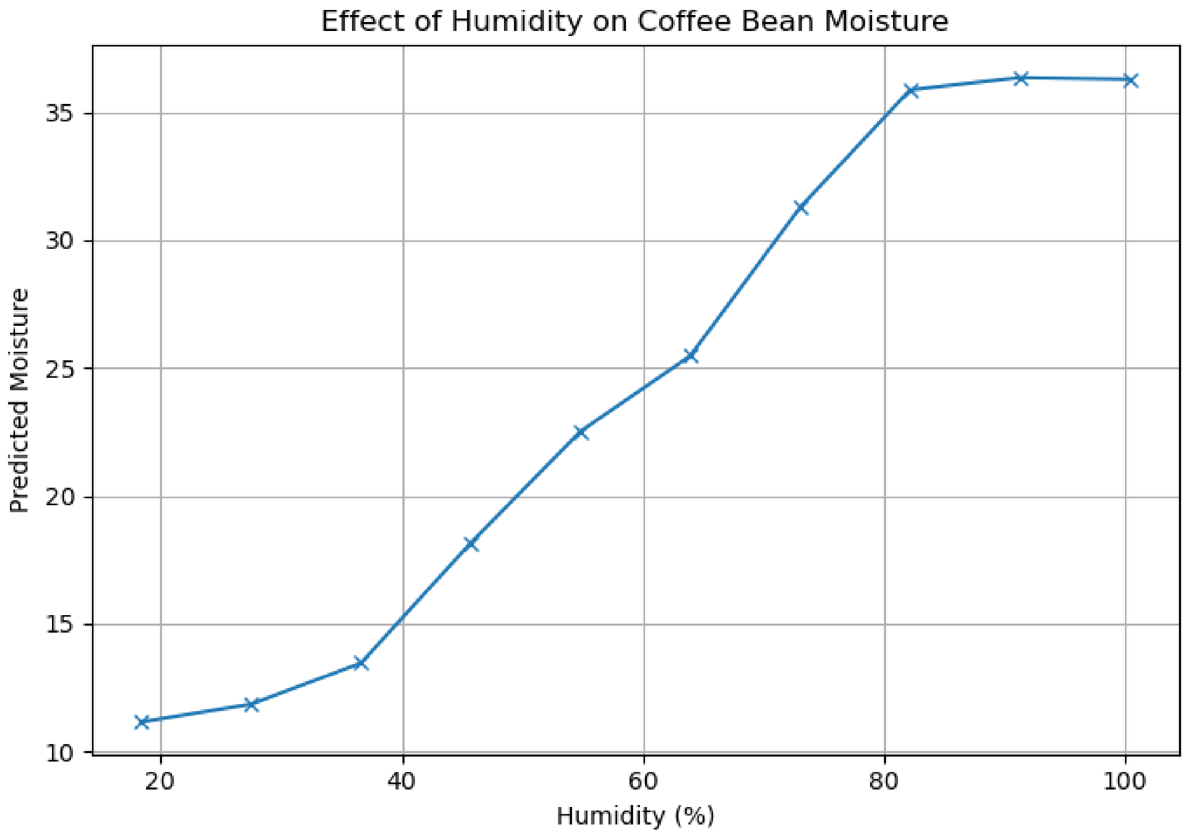
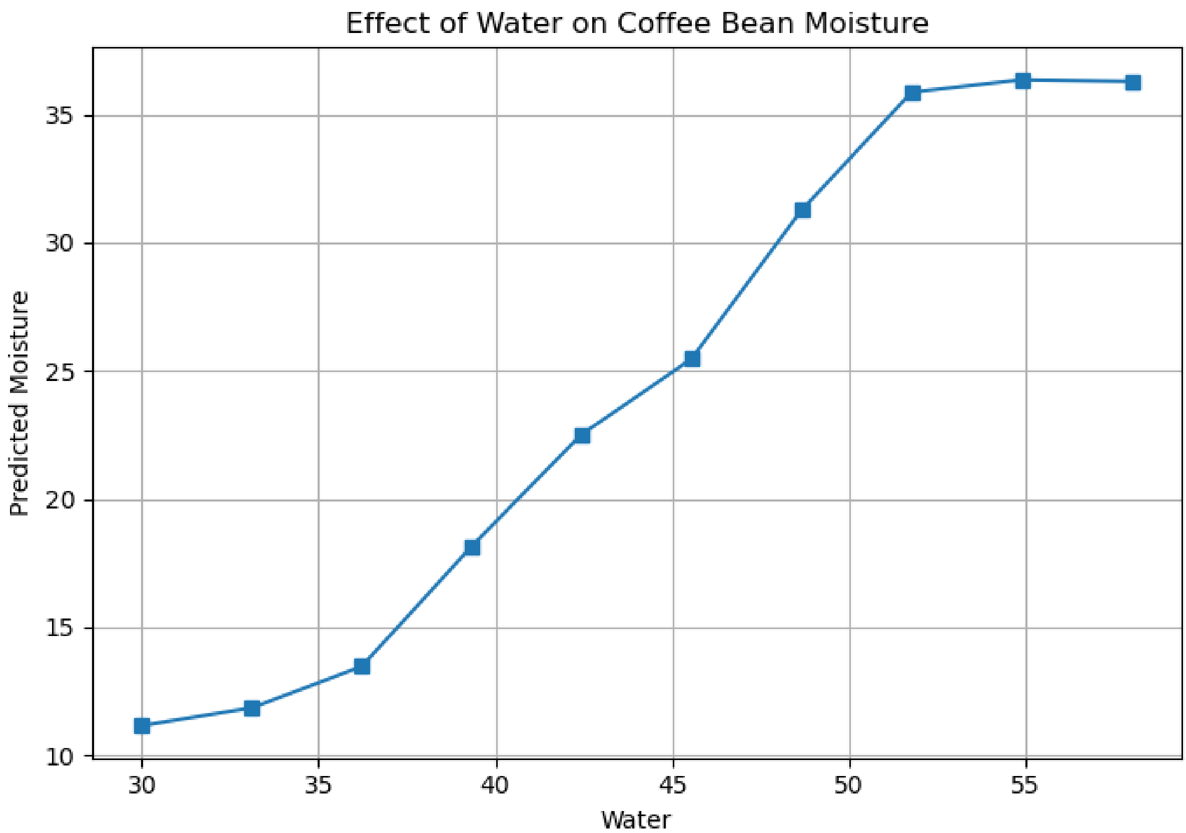
	Metric	Value
0	MSE	0.0371
1	RMSE	0.1926
2	MAE	0.1561
3	R^2	0.3263

Random Forest



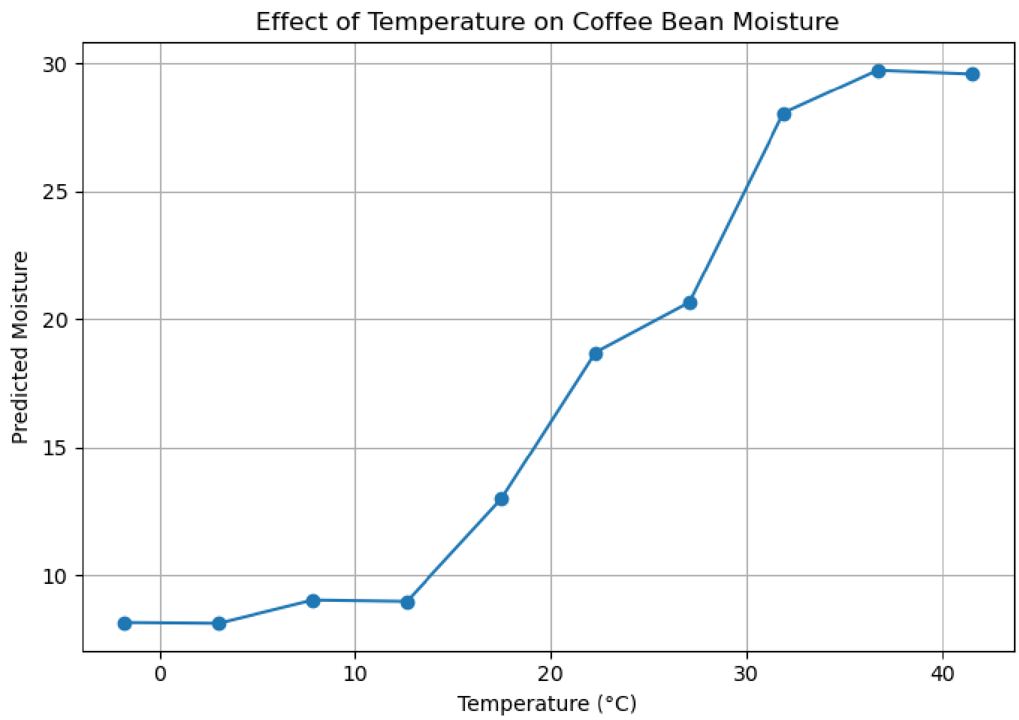

Effect of Humidity on Coffee Bean Moisture
Predicted Moisture
Humidity (%)
10
15
20
25
30
35
20
40
60
80
100

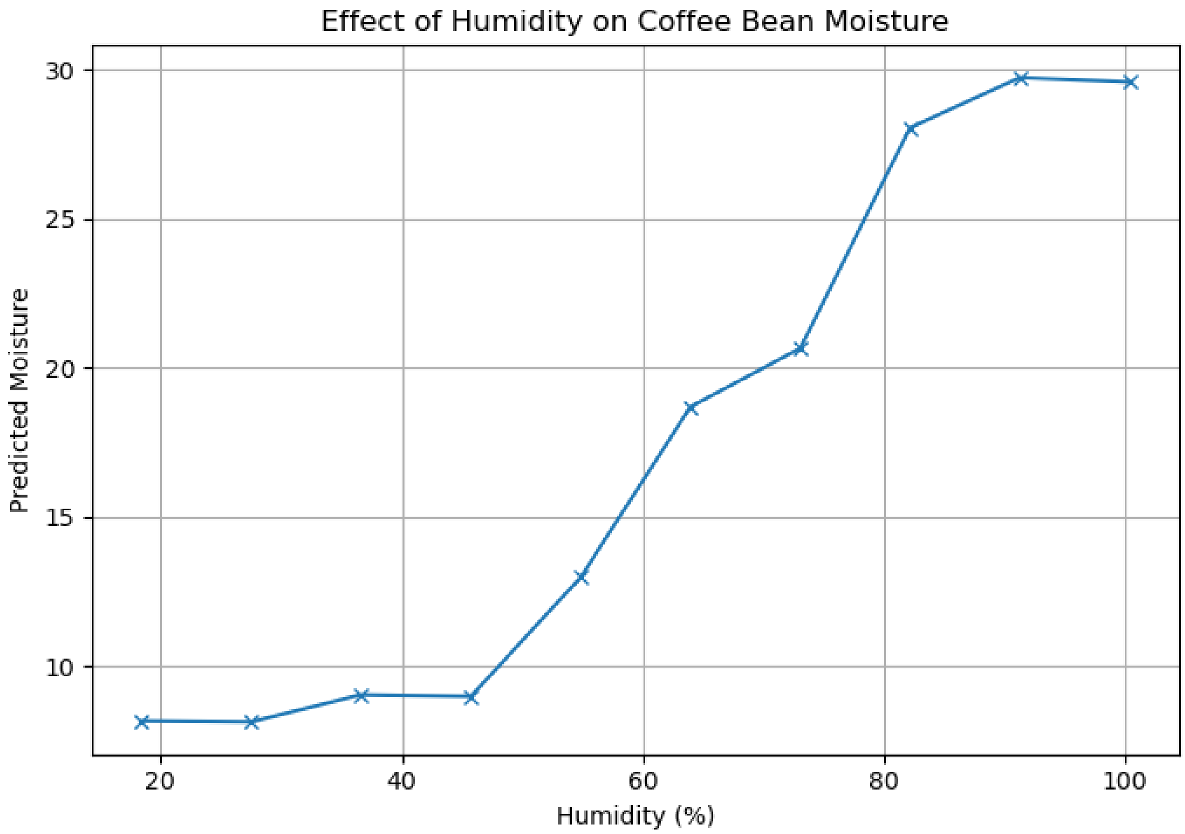

Effect of Water on Coffee Bean Moisture
Predicted Moisture
Water
10
15
20
25
30
35
30
35
40
45
50
55

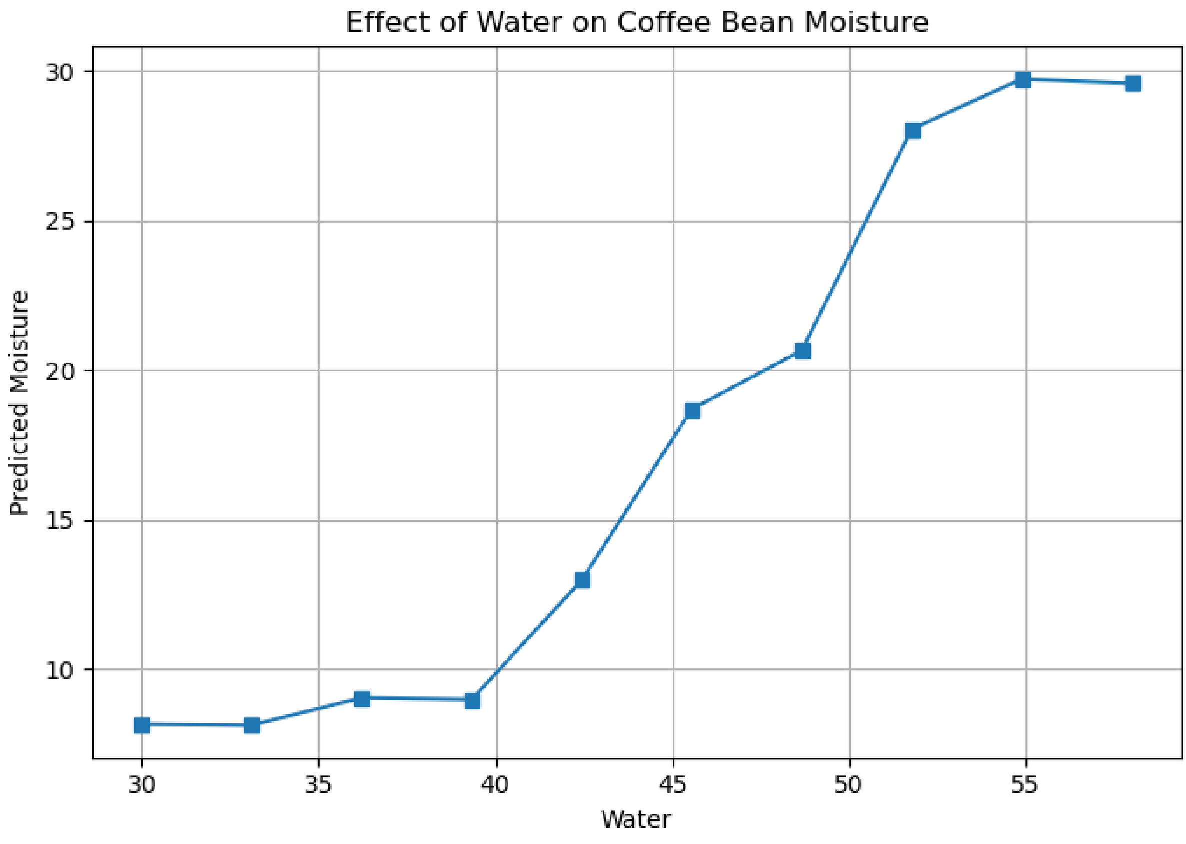
```
Model Performance on Test Set:
```

	Metric	Value
0	MSE	0.0371
1	RMSE	0.1926
2	MAE	0.1561
3	R^2	0.3263

XGBoost




Effect of Humidity on Coffee Bean Moisture
Predicted Moisture
Humidity (%)
30
25
20
15
10
20
40
60
80
100


Effect of Water on Coffee Bean Moisture
Predicted Moisture
Water
30
25
20
15
10
30
35
40
45
50
55

```
Model Performance on Test Set:
```

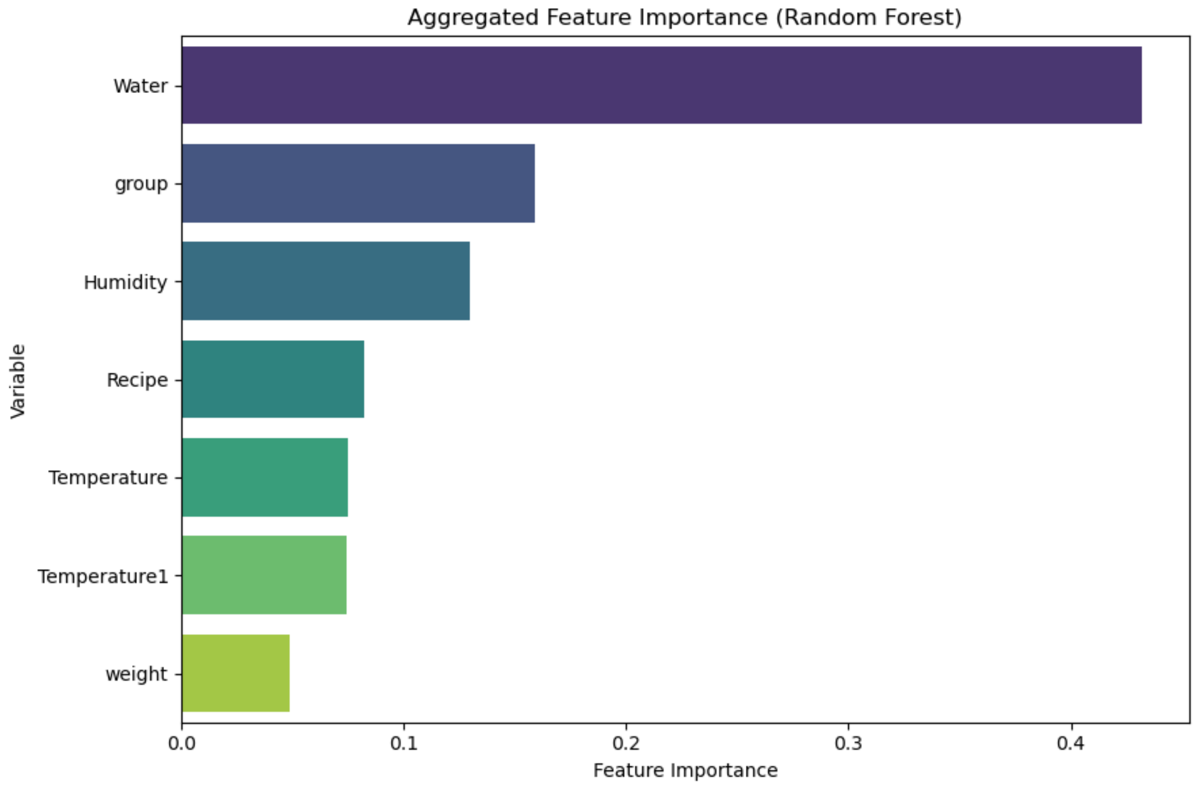
	Metric	Value
0	MSE	0.0331
1	RMSE	0.1819
2	MAE	0.1483
3	R^2	0.3990

Looking at the Machine Learning models above, XGBoost performs better in predicting the moisture content across the variables. The variables that are changing are the: Humidity, Temperature and Water. As these variables increase, the moisture content also increases indicating that the moisture content is highly dependent on these variables. However, there are some variables that are missing in this dataset, and this is the reason of why we are achieving poor performances with the machine learning models. Although XGBoost showed the strongest performance compared with the other models, the overall predictive performance remains limited because the available dataset does not fully capture the complex physical and environmental factors that influence moisture loss during roasting. Moisture content is affected by several interacting conditions, such as bean temperature, air temperature, roasting time, airflow, drum speed, pressure, initial bean moisture, bean origin, bean density, and heat transfer conditions. Since many of these variables are not included in the dataset, the models have limited information to learn from, which reduces their ability to explain the variation in moisture content accurately.

If the variables were available, the models could have performed better in predicting the moisture content during the roasting phase. This is because machine learning models depend strongly on the quality and completeness of the input features. When important explanatory variables are missing, the models may only detect partial patterns instead of the full relationship between the roasting conditions and the target variable. For example, the roasting phase is a dynamic process where moisture reduction is not only linked to water content, but also to heat exposure, temperature changes, timing, and the interaction between the beans and the roasting environment. Including additional process variables would allow the models to capture these relationships more effectively and reduce unexplained error.

Therefore, the relatively weak model performance should not be interpreted only as a limitation of the algorithms, but also as an indication that the dataset may not contain enough informative variables to represent the moisture dynamics. In future work, collecting more detailed roasting data and including additional operational variables could improve the robustness, accuracy, and practical usefulness of the predictive models.

Feature Importance



The dataset has limited variables in the dataset to predict moisture content levels. It only consists of 8 variables and 1 variable (Water), is the highly ranked in the feature importance in predicting moisture content. This shows that, the dataset is highly dependent on water as the main driver in predicting moisture content levels. Some variables interactions, are missing from the dataset and having those interactions, will improve the performance for the models.

This finding indicates that the available predictors do not fully represent the complexity of the roasting process. Moisture content during roasting is influenced by several factors that may act together rather than independently. Therefore, when the model ranks water as the most important feature, it suggests that water provides the strongest direct signal for predicting moisture content, but it also shows that the model has limited alternative information to explain changes in moisture levels. As a result, the model may rely too heavily on one dominant variable instead of learning from a wider combination of roasting conditions.

The absence of additional variables and interaction effects can reduce the ability of machine learning models to capture the real behaviour of moisture loss. In a roasting process, moisture content may be affected by the combined influence of temperature, time, humidity, air flow, bean weight, recipe, batch group, and initial moisture conditions. For example, the effect of water on moisture content may change depending on the roasting temperature or the duration of the roasting phase. Similarly, the relationship between batch group and moisture content may depend on recipe settings or environmental conditions. If these interaction effects are not included or are not represented clearly in the dataset, the models may only identify simple relationships and fail to capture more complex patterns. This can lead to weaker predictive performance, especially when the target variable is controlled by multiple physical and operational factors.

This also explains why the feature importance results should be interpreted carefully. The high importance of water does not necessarily mean that other variables are unimportant in the actual roasting process. Instead, it may reflect the limited structure of the dataset and the absence of other informative predictors. If more process variables were available, the relative importance of water could change because the model would have access to additional information explaining moisture variation. Overall, the results suggest that future work should focus on enriching the dataset with more relevant variables and interaction features. This would help reduce the model's dependence on a single dominant predictor and improve the accuracy, robustness, and practical usefulness of moisture content prediction.

Acknowledgement

I would like to take this opportunity to thank Nestle in giving me this project and using it as a final project for the CAS in Applied Data Science program at University of Bern. Upon discussing the results with the Data Science Manager from Nestle, the results are feasible as it matches the requirements of the project. However, feature engineering was required to improve the performance of the models, and more variables was required to see the interactions.

I would also like to thank University of Bern, for teaching me the concepts in Data Science and how to apply these concepts into real-world projects.

Conclusion

In conclusion, this project has taught me how to identify and capture meaningful relationships within a dataset, and how to analyse the available data to draw valid and meaningful conclusions. Through this process, I developed a better understanding of how different variables interact with one another and how these interactions can be interpreted to explain real-world outcomes. This experience strengthened my ability to think critically about data rather than simply observing it.

The project also helped me understand how to build predictive machine learning models, particularly in estimating a target variable such as moisture content under environmental and processing conditions. By using relevant features that influence these conditions, I learned how data-driven models can be used to make informed predictions. This has improved my understanding of how analytical techniques can be applied in practical settings to support decision-making and optimise processes.

References

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2. GeeksforGeeks. “What Is Exploratory Data Analysis?” *GeeksforGeeks*, 22 July 2021, www.geeksforgeeks.org/data-analysis/what-is-exploratory-data-analysis/.